NEET UG CHEMISTRY | NCERT CLASS 11 UNIT 9 | BILINGUAL / द्विभाषी

Hydrocarbons: NEET UG Master Tutorial

हाइड्रोकार्बन: नीट (NEET UG) परीक्षा हेतु सम्पूर्ण एवं प्रामाणिक ट्यूटोरियल

1. Classification of Hydrocarbons / हाइड्रोकार्बन का वर्गीकरण

Hydrocarbons are binary organic compounds containing exclusively carbon and hydrogen atoms. They are classified into:

- **Acyclic (Aliphatic):**
  - *Saturated:* Alkanes (C_nH_{2n+2}) possessing only single $C-C$ and $C-H$ σ-bonds.
  - *Unsaturated:* Alkenes (C_nH_{2n}, containing $C=C$) and Alkynes (C_nH_{2n-2}, containing $C\equiv C$).
- **Cyclic:**
  - *Alicyclic:* Cycloalkanes and cycloalkenes possessing aliphatic chemical properties.
  - *Aromatic (Arenes):* Highly stable, planar, conjugated cyclic rings adhering to Hückel's $(4n+2)\pi$ electron rule (e.g., benzene, toluene).

हिंदी सारांश: हाइड्रोकार्बन केवल कार्बन तथा हाइड्रोजन से निर्मित यौगिक हैं। ये अचक्रीय (संतृप्त ऐल्केन तथा असंतृप्त ऐल्कीन व ऐल्काइन) और चक्रीय (ऐलीसाइक्लिक तथा हकल नियम का पालन करने वाले ऐरोमैटिक एरीन) में वर्गीकृत होते हैं।

2. Alkanes (Paraffins) / ऐल्केन (पैराफिन)

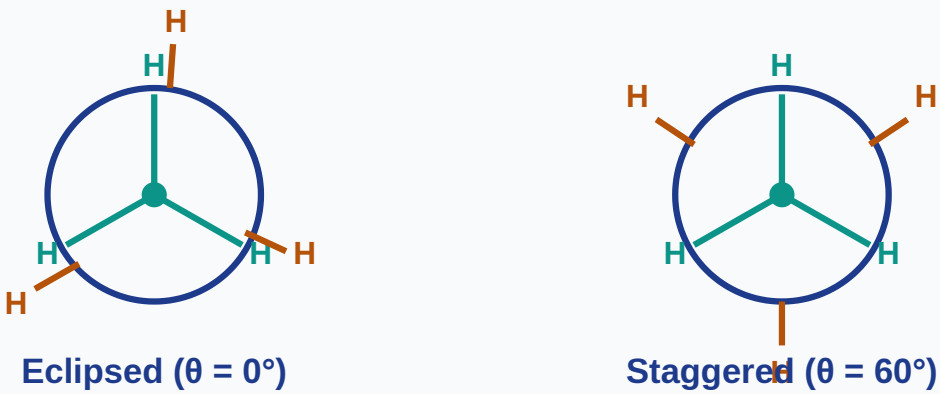
Alkanes (C_nH_{2n+2}) are saturated hydrocarbons characterised by inertness toward polar reagents under ambient conditions (hence named *paraffins*, meaning little affinity).

Structure, Bonding & Conformations

All carbon atoms are sp^3 hybridised with tetrahedral geometry (bond angle 109.5°). The $C-C$ single bond length is $154\,\text{pm}$ ($348\,\text{kJ mol}^{-1}$) and $C-H$ is $112\,\text{pm}$ ($414\,\text{kJ mol}^{-1}$).

Conformations of Ethane: Rotation about the $C-C$ σ-bond is hindered by an energy barrier of $12.55\,\text{kJ mol}^{-1}$ ($3.0\,\text{kcal mol}^{-1}$) termed **torsional strain**, caused by repulsive interactions between adjacent $C-H$ bond electron clouds.

- **Eclipsed Conformation:** Hydrogens align directly opposite each other ($\theta = 0^\circ$). Maximum torsional strain, least stable.
- **Staggered Conformation:** Hydrogens are positioned maximally far apart ($\theta = 60^\circ$). Minimum repulsion, most stable (by $12.55\,\text{kJ mol}^{-1}$).
- **Stability Order:** $\text{Staggered} > \text{Skew} > \text{Eclipsed}$. Thermal energy at room temperature rapidly interconverts conformers.



Structure Overview: Newman projections of ethane. Staggered is more stable by $12.55\,\text{kJ mol}^{-1}$.

BOSONANCE

संरचना एवं संरूपण: एथेन में एकल बंध के चारों ओर घूर्णन मरोड़ी प्रतिबल ($12.55\,\text{kJ mol}^{-1}$) के कारण बाधित होता है। सांतरित (Staggered, कोण 60°) रूप ग्रस्त (Eclipsed, कोण 0°) से अधिक स्थायी होता है।

Physical Properties

Boiling Point: Increases with molecular mass (stronger dispersion forces). As branching increases, the molecular shape becomes spherical and surface area decreases, leading to lower boiling points: n-pentane $>$ isopentane $>$ neopentane.

Melting Point: Neopentane has an unusually high melting point ($256\,\text{K}$) compared to isopentane ($113\,\text{K}$) due to close packing of spherical molecules in the crystal lattice.

Preparation Methods

1. **Catalytic Hydrogenation:** $R-CH=CH_2 + H_2 \xrightarrow{Pt/Pd \text{ or } Ni,\ \Delta} R-CH_2-CH_3$
2. **Reduction of Alkyl Halides:** $R-X + Zn + H^+ \longrightarrow R-H + Zn^{2+} + X^-$
3. **Wurtz Reaction:** $2R-X + 2Na \xrightarrow{\text{dry ether}} R-R + 2NaX$ (Best for symmetrical alkanes with even carbon count; two different halides yield a 3-alkane mixture; methane cannot be made).
4. **Decarboxylation (Dumas):** $R-COONa + NaOH \xrightarrow{CaO,\ \Delta} R-H + Na_2CO_3$ (Alkane has one less carbon).
5. **Kolbe's Electrolysis:**

   $2RCOO^-Na^+ + 2H_2O \xrightarrow{\text{electrolysis}} R-R + 2CO_2\,(\text{Anode}) + H_2\,(\text{Cathode}) + 2NaOH$ (Solution becomes basic, pH rises).

Chemical Properties

- **Free Radical Chlorination Mechanism:**
  1. *Initiation:* $Cl_2 \xrightarrow{h\nu} 2\,Cl^\bullet$ (Homolytic bond fission).
  2. *Propagation:* $CH_4 + Cl^\bullet \longrightarrow CH_3^\bullet + HCl$ (Rate-determining step); $CH_3^\bullet + Cl_2 \longrightarrow CH_3Cl + Cl^\bullet$.
  3. *Termination:* Coupling of radicals ($Cl^\bullet + Cl^\bullet \rightarrow Cl_2$; $CH_3^\bullet + Cl^\bullet \rightarrow CH_3Cl$; $CH_3^\bullet + CH_3^\bullet \rightarrow C_2H_6$). Formation of trace ethane confirms methyl free radicals.
- **Controlled Oxidation:**
  - $2CH_4 + O_2 \xrightarrow{Cu\ /\ 523\,\text{K},\ 100\,\text{atm}} 2CH_3OH$
  - $CH_4 + O_2 \xrightarrow{Mo_2O_3,\ \Delta} HCHO + H_2O$
  - $(CH_3)_3CH \xrightarrow{[O],\ KMnO_4} (CH_3)_3C-OH$ (Selective oxidation of tertiary hydrogen).
- **Aromatisation:** $CH_3(CH_2)_4CH_3 \xrightarrow[773\,\text{K},\ 10-20\,\text{atm}]{Cr_2O_3\ /\ Al_2O_3} C_6H_6\,(\text{Benzene}) + 4H_2$

3. Alkenes (Olefins) / ऐल्कीन

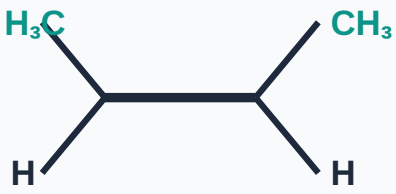
Alkenes (C_nH_{2n}) possess a carbon-carbon double bond ($C=C$) consisting of one strong σ-bond ($397\,\text{kJ mol}^{-1}$) and one lateral π-bond ($284\,\text{kJ mol}^{-1}$). The $C=C$ bond length is $134\,\text{pm}$. The exposed, loosely held π-electrons act as a nucleophile, directing alkenes predominantly toward **electrophilic addition reactions**.

Geometrical Isomerism (cis-trans)

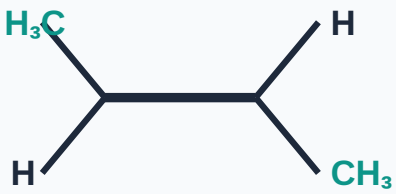
Restricted rotation around the $C=C$ π-bond produces stereoisomers in alkenes of the form $RCH=CHR'$:

- **cis-isomer:** Identical groups on the same side. Vector dipoles reinforce ($\mu > 0$), giving higher polarity and a **higher boiling point**.
- **trans-isomer:** Identical groups on opposite sides. Vector dipoles oppose ($\mu \approx 0$). Symmetrical packing in the crystal lattice gives it a **higher melting point** and lower solubility than the cis-isomer.

BOSONANCE



cis-But-2-ene (μ = 0.33 D, Higher b.p.)



trans-But-2-ene (μ = 0, Higher m.p.)

Stereochemical Comparison: Geometrical isomers of but-2-ene and dipole moments.

Preparation of Alkenes

- **From Alkynes:**
  - *cis-alkene:* Hydrogenation over **Lindlar's catalyst** ($H_2, Pd/BaSO_4 + \text{quinoline}$).
  - *trans-alkene:* Birch reduction with Na **in liquid** NH_3 at $195\,\text{K}$.
- **Dehydrohalogenation of Alkyl Halides (β-Elimination):** Heating with alc. KOH. Follows **Saytzeff's Rule:** The more substituted (hyperconjugatively stabilised) alkene is the major product.
- **Dehalogenation of Vicinal Dihalides:** $BrCH_2 - CH_2Br + Zn \xrightarrow{\Delta} CH_2 = CH_2 + ZnBr_2$.

Electrophilic Addition Reactions

1. Markovnikov's Rule

In the addition of an unsymmetrical reagent (HX) to an unsymmetrical alkene, the negative part attaches to the doubly bonded carbon carrying fewer hydrogens. Electrophilic addition begins with H^+ addition to yield the more stable carbocation intermediate ($3^\circ > 2^\circ > 1^\circ$):

$$CH_3 - CH = CH_2 + H^+ \longrightarrow CH_3 - \overset{+}{C}H - CH_3 \ (\text{stable } 2^\circ \text{ carbocation}) \xrightarrow{Br^-} CH_3 - CH(Br) - CH_3$$

2. Kharasch Peroxide Effect (Anti-Markovnikov Addition)

In the presence of organic peroxides (e.g., benzoyl peroxide), HBr **only** adds contrary to Markovnikov's rule, generating 1-bromopropane via a free-radical chain mechanism.

> **Why only HBr? (Thermodynamic Explanation)**
>
> Both propagation steps (addition of $X^\bullet$ to the alkene and abstraction of H from $H - X$) must be exothermic ($\Delta H < 0$). For HCl, the $H - Cl$ bond is too strong ($430.5\,\text{kJ mol}^{-1}$), making the second step endothermic. For HI, the $H - I$ bond is weak, but $I^\bullet$ radicals rapidly couple into I_2 ($2I^\bullet \rightarrow I_2$), making the first step endothermic. **Only for HBr are both steps exothermic.** Hence, HCl and HI do not exhibit the peroxide effect.

3. Baeyer's Test & Ozonolysis

- **Baeyer's Test:** Decolourisation of cold, dilute neutral/alkaline $KMnO_4$ (pink to brown MnO_2) with the formation of vicinal glycols tests for unsaturation:

$$CH_2 = CH_2 + H_2O + [O] \xrightarrow{\text{cold dil. } KMnO_4} CH_2(OH) - CH_2(OH)$$

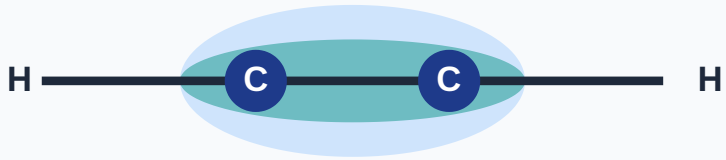
- **Ozonolysis:** Cleavage with O_3 followed by Zn/H_2O identifies double bond position:

$$R_2C = CHR' \xrightarrow{O_3} \text{Ozonide} \xrightarrow{Zn\ /\ H_2O} R_2C = O + R'CHO + ZnO$$

4. Alkynes / ऐल्काइन

Alkynes (C_nH_{2n-2}) contain a $C \equiv C$ triple bond comprising one coaxial σ-bond and two mutually perpendicular π-bonds, with linear geometry (180°). The $C \equiv C$ bond length is $120\,\text{pm}$ ($823\,\text{kJ mol}^{-1}$).

BOSONANCE



Linear sp (180°), Cylindrical π-cloud (120 pm)

Orbital Representation: Cylindrical π-electron sheath and linear geometry of ethyne.

Preparation Methods

1. **From Calcium Carbide:** $CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + HC \equiv CH \uparrow$
2. **From Vicinal Dihalides:**

$$CH_2Br - CH_2Br \xrightarrow{\text{alc. } KOH} CH_2 = CHBr \xrightarrow{NaNH_2} HC \equiv CH$$

Note: Sodamide ($NaNH_2$) is required in the second step to eliminate HBr from the unreactive vinyl halide.

Acidic Character of Terminal Alkynes

The sp hybridised carbon in alkynes has 50% s-character, rendering it significantly more electronegative than sp^2 (33.3%) or sp^3 (25%) carbon. Consequently, the $C - H$ bond in terminal alkynes is strongly polarised, and hydrogen can be abstracted by strong bases like Na metal or $NaNH_2$:

$$HC \equiv CH + Na \longrightarrow HC \equiv C^-Na^+ + \frac{1}{2}H_2 \uparrow$$

Relative Acidity Order: $HC \equiv CH > CH_3 - C \equiv CH \gg CH_2 = CH_2 > CH_3 - CH_3$
$H_2O > ROH > HC \equiv CH > NH_3 > CH_4$

Key Reactions of Alkynes

- **Hydration (Kucherov Reaction):** Alkynes add water in the presence of $dil.\ H_2SO_4 + HgSO_4$ at $333\ \text{K}$. Ethyne yields **ethanal**; all higher terminal and internal alkynes yield **ketones**:

$$HC \equiv CH + H_2O \xrightarrow[333\ \text{K}]{HgSO_4\ /\ H_2SO_4} [CH_2 = CH - OH] \xrightarrow{\text{tautomerises}} CH_3 - CHO$$

$$CH_3 - C \equiv CH + H_2O \xrightarrow{Hg^{2+}\ /\ H^+} [CH_3 - C(OH) = CH_2] \xrightarrow{\text{tautomerises}} CH_3 - CO - CH_3$$

- **Cyclic Trimerisation:** Passing ethyne through a red-hot iron tube at $873\ \text{K}$ synthesises benzene:

$$3\,HC \equiv CH \xrightarrow{\text{Red hot Fe tube, 873 K}} C_6H_6\ (\text{Benzene})$$

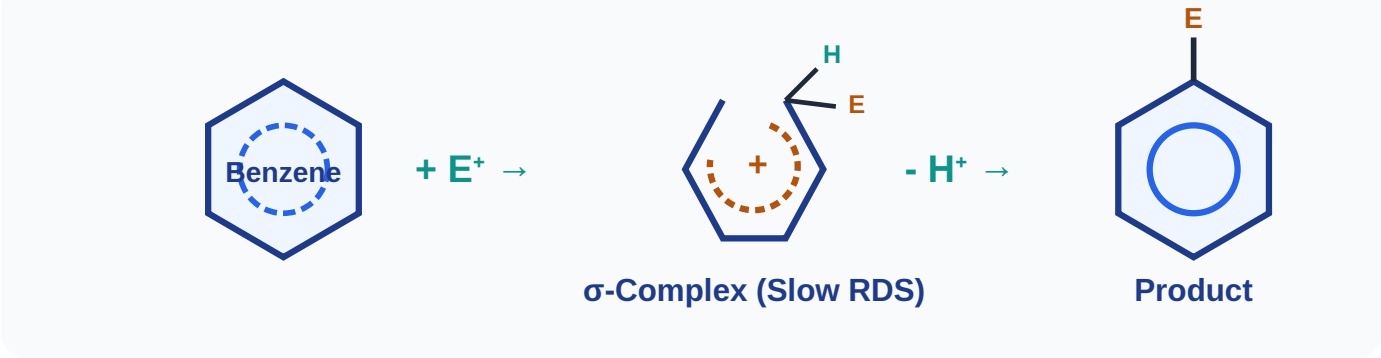
5. Aromatic Hydrocarbons (Arenes) / ऐरोमैटिक हाइड्रोकार्बन

Benzene (C_6H_6) is a planar regular hexagonal molecule where all six $C - C$ bonds are equal ($139\ \text{pm}$), intermediate between single ($154\ \text{pm}$) and double ($134\ \text{pm}$) bonds. The resonance stabilisation energy of benzene is $150.6\ \text{kJ mol}^{-1}$ ($36\ \text{kcal mol}^{-1}$).

Aromaticity (Hückel's Rule)

A planar, cyclic, conjugated ring system possessing $(4n + 2)\pi$ **electrons** ($n = 0, 1, 2, 3, \ldots$) is aromatic. Benzene has 6π electrons ($n = 1$); cyclooctatetraene has 8π electrons ($4n$) and avoids antiaromaticity by adopting a non-planar **tub-shaped** conformation, making it non-aromatic.

BOSONANCE



Reaction Pathway: Three-step $S_E\mathrm{Ar}$ mechanism of benzene via the arenium ion (σ-complex).

Electrophilic Aromatic Substitution ($S_E\mathrm{Ar}$) Mechanism

1. **Generation of Electrophile (E^+):**
   - *Nitration:* $HNO_3 + 2H_2SO_4 \rightleftharpoons NO_2^+ + H_3O^+ + 2HSO_4^-$ (Electrophile: NO_2^+).
   - *Halogenation:* $Cl_2 + AlCl_3 \rightleftharpoons Cl^+ + [AlCl_4]^-$ (Electrophile: Cl^+).
   - *Sulphonation:* $2H_2SO_4 \rightleftharpoons SO_3 + H_3O^+ + HSO_4^-$ (Neutral electrophile: SO_3).
2. **Formation of Arenium Ion (σ-Complex / Wheland Intermediate):** Electrophile attacks the ring, disrupting aromaticity and forming an sp^3 hybridised carbon. The carbocation is resonance-stabilised across ortho and para carbons. **This is the slow, rate-determining step.**
3. **Loss of Proton:** Fast abstraction of the proton from the sp^3 carbon by a base restores the aromatic sextet.

Friedel-Crafts Reactions & Directive Influence

Alkylation: $C_6H_6 + CH_3Cl \xrightarrow{\text{anhyd. } AlCl_3} C_6H_5-CH_3 + HCl$. When 1-chloropropane is used, the 1° propyl carbocation rearranges via a **1,2-hydride shift** to the 2° isopropyl carbocation; the major product is **cumene (isopropylbenzene)**.

Acylation: $C_6H_6 + CH_3COCl \xrightarrow{\text{anhyd. } AlCl_3} C_6H_5-COCH_3 + HCl$. The acylium ion ($CH_3-\overset{+}{C}=O$) is resonance-stabilised; hence, **no rearrangement occurs**.

Group Type	Electronic Effect	Ring Reactivity	Directive Influence	Examples
Activating	$+M/+R$ or $+I$	Activated	**Ortho and Para**	$-OH, -NH_2, -OCH_3, -CH_3$
Deactivating	$-M/-R$ and $-I$	Deactivated	**Meta**	$-NO_2, -CN, -CHO, -COOH$
Halogens (Exception)	Strong $-I$ > Weak $+R$	**Deactivated**	**Ortho and Para**	$-F, -Cl, -Br, -I$

Carcinogenicity and Toxicity

Polynuclear aromatic hydrocarbons (PAHs) with more than two fused benzene rings (e.g., 1,2-benzanthracene, 3-methylcholanthrene, 1,2-benzpyrene) are formed by incomplete combustion of coal, petroleum, and tobacco smoke. In vivo, they are metabolised into reactive diol-epoxides that covalently bind to DNA guanine bases, inducing mutations and oncogenesis.

BOSONANCE

6. High-Yield Worked Examples / हल किए गए उदाहरण

Example 1: Alkene Ozonolysis Deduction / ओजोनीकरण द्वारा संरचना निर्धारण

Problem: An alkene 'A' on reductive ozonolysis (O_3, then Zn/H_2O) produces propanone and ethanal in 1:1 molar ratio. Determine 'A' and state if it exhibits geometrical isomerism.

Solution:

Align the two carbonyl products: $(CH_3)_2C=O$ and $O=CH-CH_3$. Remove oxygens and connect carbons: $(CH_3)_2C=CH-CH_3$ (**2-Methylbut-2-ene**). Carbon-2 holds two identical methyl groups; therefore, **it cannot exhibit geometrical isomerism**.

Example 2: Friedel-Crafts Carbocation Rearrangement / फ्रीडेल-क्राफ्ट्स पुनर्विन्यास

Problem: Predict the major product when benzene is treated with 1-chloropropane in the presence of anhydrous $AlCl_3$.

Solution:

1-Chloropropane reacts with $AlCl_3$ to form a 1° carbocation ($CH_3CH_2\overset{+}{C}H_2$), which spontaneously undergoes a **1,2-hydride shift** to generate the more stable 2° isopropyl carbocation ($CH_3-\overset{+}{C}H-CH_3$). Electrophilic attack on benzene yields **cumene (isopropylbenzene)** as the major product.

Example 3: Aromaticity Verification / ऐरोमैटिकता का निर्धारण

Problem: Classify (i) Cyclopentadienyl anion and (ii) Cyclooctatetraene as aromatic, antiaromatic, or non-aromatic.

Solution:

(i) *Cyclopentadienyl anion:* Cyclic, planar, conjugated with two π-bonds and one negative-charge lone pair $= 6\pi$ electrons. Matches $4n+2$ $(n=1) \implies$ **Aromatic**.
(ii) *Cyclooctatetraene:* Contains 8π electrons $(4n)$. To avoid antiaromatic destabilisation, it puckers into a non-planar **tub conformation** $\implies$ **Non-aromatic**.

BOSONANCE

7. NEET UG Practice Questions / नीट (NEET UG) अभ्यास प्रश्न

Practice Question 1 (NEET Single-Correct MCQ)

Which of the following alkanes cannot be synthesised in good yield by the Wurtz reaction?

(A) n-Hexane

(B) 2,3-Dimethylbutane

(C) n-Heptane

(D) n-Butane

Correct Answer: (C) n-Heptane

Explanation: The Wurtz reaction gives high yields for symmetrical alkanes with an even number of carbons. Synthesising an odd-carbon alkane like n-heptane requires two different alkyl halides (C_3 and C_4), which produces an inseparable mixture of three alkanes ($C_6H_{14}, C_7H_{16}, C_8H_{18}$).

अभ्यास प्रश्न 1 (हिंदी रूपांतरण)

निम्नलिखित में से किस ऐल्केन को वुर्ट्ज़ अभिक्रिया द्वारा उत्तम लब्धि में नहीं बनाया जा सकता है?

(A) n-हेक्सेन

(B) 2,3-डाइमेथिलब्यूटेन

(C) n-हेप्टेन

(D) n-ब्यूटेन

सही उत्तर: (C) n-हेप्टेन

विस्तृत हल: विषम कार्बन संख्या वाले ऐल्केन (n-हेप्टेन) बनाने के लिए दो भिन्न ऐल्किल हैलाइड लेने पड़ते हैं, जिससे तीन ऐल्केनों का जटिल मिश्रण प्राप्त होता है जिसे पृथक करना कठिन होता है।

BOSONANCE

Practice Question 2 (NEET Assertion-Reason)

Assertion (A): Chlorobenzene is less reactive than benzene toward electrophilic aromatic substitution, but directs incoming electrophiles to ortho and para positions.
Reason (R): The strong $-I$ effect of chlorine deactivates the ring overall, but its $+R$ resonance effect selectively stabilises the arenium ion formed at ortho and para positions.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

Explanation: In halobenzenes, inductive withdrawal ($-I$) outweighs resonance donation ($+R$), decreasing net ring electron density (deactivation). However, during ortho/para attack, chlorine's lone pair directly participates in resonance, lowering activation energy for those pathways.

अभ्यास प्रश्न 2 (हिंदी रूपांतरण)

अभिकथन (A): क्लोरोबेंजीन, इलेक्ट्रॉनरागी प्रतिस्थापन के प्रति बेंजीन से कम क्रियाशील है परन्तु यह ऑर्थो-पैरा निर्देशक है।
कारण (R): क्लोरीन का प्रबल $-I$ प्रभाव वलय को निष्क्रिय करता है, जबकि $+R$ अनुनाद प्रभाव ऑर्थो-पैरा एरेनियम आयन को स्थायित्व देता है।

(A) (A) और (R) दोनों सही हैं तथा (R), (A) की सही व्याख्या है।

(B) (A) और (R) दोनों सही हैं परन्तु (R), (A) की सही व्याख्या नहीं है।

(C) (A) सही है परन्तु (R) गलत है।

(D) (A) गलत है परन्तु (R) सही है।

सही उत्तर: (A) (A) और (R) दोनों सही हैं तथा (R), (A) की सही व्याख्या है।

विस्तृत हल: क्लोरीन में $-I > +R$ होने से वलय निष्क्रिय होता है, परन्तु ऑर्थो तथा पैरा आक्रमण में अनुनादी स्थायित्व मिलने के कारण यह ऑर्थो-पैरा निर्देशक होता है।

BOSONANCE

Practice Question 3 (NEET Statement-Based MCQ)

Statement I: Addition of HBr to propene in the presence of benzoyl peroxide yields 1-bromopropane.
Statement II: Under identical conditions, HCl and HI also give anti-Markovnikov addition products.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Correct Answer: (C) Statement I is correct but Statement II is incorrect.

Explanation: The peroxide effect is observed exclusively with HBr. For HCl, the $H-Cl$ bond is too strong for radical propagation. For HI, iodine radicals recombine into I_2 rather than adding to the alkene. Hence, HCl and HI always add via the Markovnikov rule.

अभ्यास प्रश्न 3 (हिंदी रूपांतरण)

कथन I: परॉक्साइड की उपस्थिति में प्रोपीन पर HBr का योग 1-ब्रोमोप्रोपेन देता है।
कथन II: समान परिस्थितियों में HCl तथा HI भी ऐन्टी-मार्कोनीकॉफ योगज उत्पाद देते हैं।

(A) कथन I और II दोनों सही हैं।

(B) कथन I और II दोनों गलत हैं।

(C) कथन I सही है परन्तु कथन II गलत है।

(D) कथन I गलत है परन्तु कथन II सही है।

सही उत्तर: (C) कथन I सही है परन्तु कथन II गलत है।

विस्तृत हल: परॉक्साइड प्रभाव केवल HBr द्वारा दर्शाया जाता है। उष्मागतिकीय कारणों से HCl तथा HI मार्कोनीकॉफ नियम का ही पालन करते हैं।

BOSONANCE

Practice Question 4 (NEET Reaction Sequence & Matching)

Match the transformations in Column I with the reagents in Column II:

Column I (Transformation)	Column II (Reagent)
(a) But-2-yne $\rightarrow$ cis-But-2-ene	(i) $dil.\ H_2SO_4 + HgSO_4,\ 333\,\text{K}$
(b) But-2-yne $\rightarrow$ trans-But-2-ene	(ii) $Cr_2O_3/Al_2O_3,\ 773\,\text{K}$
(c) Propyne $\rightarrow$ Propanone	(iii) $H_2,\ Pd/BaSO_4 + \text{quinoline}$
(d) n-Hexane $\rightarrow$ Benzene	(iv) Na in liquid NH_3

(A) a $\rightarrow$ iii, b $\rightarrow$ iv, c $\rightarrow$ i, d $\rightarrow$ ii

(B) a $\rightarrow$ iv, b $\rightarrow$ iii, c $\rightarrow$ i, d $\rightarrow$ ii

(C) a $\rightarrow$ iii, b $\rightarrow$ iv, c $\rightarrow$ ii, d $\rightarrow$ i

(D) a $\rightarrow$ i, b $\rightarrow$ ii, c $\rightarrow$ iii, d $\rightarrow$ iv

Correct Answer: (A) a $\rightarrow$ iii, b $\rightarrow$ iv, c $\rightarrow$ i, d $\rightarrow$ ii

Explanation: Lindlar's catalyst gives cis-alkene (iii); Birch reduction gives trans-alkene (iv); Kucherov hydration of propyne gives propanone (i); Aromatisation of n-hexane uses Cr_2O_3/Al_2O_3 (ii).

अभ्यास प्रश्न 4 (हिंदी रूपांतरण)

स्तंभ I का स्तंभ II से सही मिलान कीजिए।

(A) a $\rightarrow$ iii, b $\rightarrow$ iv, c $\rightarrow$ i, d $\rightarrow$ ii

(B) a $\rightarrow$ iv, b $\rightarrow$ iii, c $\rightarrow$ i, d $\rightarrow$ ii

(C) a $\rightarrow$ iii, b $\rightarrow$ iv, c $\rightarrow$ ii, d $\rightarrow$ i

(D) a $\rightarrow$ i, b $\rightarrow$ ii, c $\rightarrow$ iii, d $\rightarrow$ iv

सही उत्तर: (A) a $\rightarrow$ iii, b $\rightarrow$ iv, c $\rightarrow$ i, d $\rightarrow$ ii

विस्तृत हल: लिंडलर उत्प्रेरक सिस-उत्पाद देता है; बर्च अपचयन ट्रांस-उत्पाद देता है; कुचेरोव अभिक्रिया कीटोन बनाती है; तथा क्रोमिया उत्प्रेरक ऐरोमैटिकरण करता है।

8. Common Pitfalls & Revision Checklist / सावधानियाँ एवं चेकलिस्ट

Frequent NEET Traps:

- **Kolbe's Gas Distribution:** Anode evolves Alkane + CO_2; Cathode evolves H_2; Solution pH becomes alkaline.
- **Terminal Alkyne Acidity:** Only terminal alkynes ($R-C\equiv CH$) react with Na or $NaNH_2$. Non-terminal alkynes (like but-2-yne) do not react.
- **Peroxide Selectivity:** Operates strictly for HBr. With HCl or HI, always apply Markovnikov's rule.
- **Halobenzene Directive Effect:** Halogens are *deactivating* yet *ortho/para-directing*.

Rapid Revision Points:

- ☑ Staggered ethane is more stable than eclipsed by $12.55\,\text{kJ mol}^{-1}$ (torsional barrier).
- ☑ Lindlar's catalyst ($Pd/BaSO_4 + \text{quinoline}$) $\rightarrow$ cis-alkene; Birch reduction ($Na/liq.\ NH_3$) $\rightarrow$ trans-alkene.
- ☑ Hydration of ethyne yields ethanal; all other alkynes yield ketones.
- ☑ Hückel's Rule requires a planar, cyclic, conjugated ring with $(4n+2)\pi$ electrons.

BOSONANCE

9. Concise Final Recap / अंतिम सारांश

Hydrocarbons form the bedrock of organic chemistry for NEET UG:

- **Alkanes:** Saturated sp^3 systems undergoing free radical substitution; boiling points decrease with branching; conformations governed by torsional strain.
- **Alkenes:** Planar sp^2 olefins exhibiting cis-trans isomerism; undergo electrophilic addition (Markovnikov) and peroxide-assisted radical addition (HBr only); cleaved by ozonolysis.
- **Alkynes:** Linear sp systems; terminal alkynes are acidic; undergo Kucherov hydration to carbonyls and cyclic trimerisation to benzene.
- **Arenes:** Aromatic $(4n+2)\pi$ rings undergoing electrophilic substitution via the arenium ion (σ-complex); halogens are deactivating yet ortho/para-directing.
- **Carcinogenicity:** Polynuclear aromatic hydrocarbons (PAHs) from incomplete combustion form covalent DNA adducts inducing cancer.

> **अंतिम सारांश:** यह ट्यूटोरियल ऐल्केन, ऐल्कीन, ऐल्काइन तथा ऐरोमैटिक यौगिकों की संरचना, विरचन, अभिक्रियाओं, इलेक्ट्रॉनिक क्रियाविधियों तथा कैंसरजन्यता का संपूर्ण NEET UG पाठ्यक्रम आच्छादित करता है।

BOSONANCE